

Development of dynamic growth and body composition models for Nile Tilapia (*Oreochromis niloticus*): An exploratory approach to protein and energy metabolism

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ABSTRACT

The aquaculture sector has been steadily growing and thus there is an increasing need to develop mathematical models that allow the estimation of production-related parameters. Prediction of fish growth, feed requirements and waste outputs, are essential in order to ensure the profitability and sustainability of the production activities. Bioenergetic models have been widely used to estimate growth based on energy budgets, but they have some limitations by not explicitly considering the mass-balance of the main macronutrients (e.g., protein). In turn, nutrient-based models are more explanatory, as they consider both energy and nutrient inputs, and estimate fish growth by simulating nutrient accumulation in the fish body. Although some bioenergetic and nutrient-based models for predicting Nile tilapia growth exist in the literature, their suitability is not entirely clear, since their development is often based on uncertain or suboptimal criteria (e.g., relying solely on calibration goodness-of-fit measures).

In this work, Nile tilapia growth datasets covering a wide range of rearing conditions and feed compositions were collected from the scientific literature. An exploratory analysis of the collected data was performed to clarify the relationships between energy/protein intake and gain. In this analysis, a direct relationship was observed between digestible energy intake and energy gain, as well as between digestible protein intake and protein gain. Protein gain showed better efficiency than energy gain, even at higher intake levels and without clear evidence of a saturation effect. While digestible energy intake negatively affects energy retention efficiency, digestible protein intake does not significantly impact protein retention efficiency. Furthermore, while energy retention efficiency varies with fish body weight, the same effect was not observed for protein retention efficiency. Finally, though DP/DE ratio has no apparent effect on energy retention efficiency, it seems to negatively affect protein retention efficiency. Considering these observations, plausible growth models with different levels of complexity were developed and calibrated under a diverse set of assumptions. Additionally, two growth models already published for Nile tilapia were calibrated using the same datasets, and their performance was compared with the models developed in the present study. The exploratory analysis of the data suggests a direct relationship between digestible energy intake and energy gain, and between digestible protein intake and protein gain. Furthermore, fish body weight seems to affect the energy retention efficiency. The results of model evaluation showed that energy-protein flux models (EP models) have lower errors in predicting fish growth than pure bioenergetic models (MAPE_{bw} ~ 9% against ~13%, respectively), showing the importance of considering protein intake when estimating Nile tilapia growth. Furthermore, assuming the fixed standard metabolic body weight exponents of 0.80 and 0.70 (for energy and protein, respectively), rather than estimating them from the data, seemingly improved the predictive ability of the models. This approach demonstrates the benefits of coupling bioenergetics with nutrient-based models to predict growth and body composition of Nile tilapia along time.

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1. Introduction

To achieve optimal performance in aquaculture, it is important to have good control of feeding by adapting it to the current state of production, in order to ensure optimal growth at reduced costs and low environmental impact. Therefore, feeding strategies should be optimized, among other considerations, through a careful daily estimation of the standing biomass. Due to the impracticality of daily measurement of fish biomass, it is crucial to understand in an objective and practical way, the growth patterns, and their underlying mechanisms, to allow (for example) biomass estimation between measurements. Mathematical models are a good tool to use in such cases, where it is impracticable to directly measure the system response, and also assist in data interpretation and decision support (Føre et al., 2018).

Traditionally, empirical mathematical models (e.g., SGR, TGC, FCR) have been used in aquaculture to predict growth due to their simplicity and ease of parameter determination. However, such models often ignore important aspects, such as feeding information (e.g., feed quantities or properties), and/or lack clear physical/biological support, being inadequate as general context-independent prediction models. In academic contexts, other alternatives have been suggested, such as the use of bioenergetic and nutrient-based models, which are based on the principles of energy and nutrient balance:

$$\text{Net balance} = \text{intake} - \text{losses}$$

These types of models take into consideration feed quantities and (at least some) properties, which allows them to provide more precise growth predictions. A common bioenergetic modelling approach is to describe the partitioning of food energy into excretory energy (FE, UE and ZE), heat production/increment (HiE) and recovered/retained energy (RE) (Brett and Groves, 1979; Xie et al., 2011). The RE can therefore be calculated as (Xie et al., 1997, 2011):

$$\text{RE} = \text{IE} - \text{FE} - (\text{UE} + \text{ZE}) - \text{HiE},$$

where IE represents the intake component, while $\text{FE} + (\text{UE} + \text{ZE}) + \text{HiE}$ represents the losses component. However, this type of model can be converted to a simpler model where the losses are subdivided into two components: intake-dependent losses ($\text{COG} \times \text{DE}$) and intake-independent losses (FM) – the advantage of this approach is that it is simpler (i.e., less parameters) and thus more identifiable and easier to calibrate:

$$\text{RE} = (1 - \text{COG}) \times \text{DE} - \text{FM}$$

where COG represents the costs of growth – in this case costs of energy deposition (Bureau et al., 2000; Conceição et al., 1998) – usually equated with the specific dynamic action (SDA), DE is the digestible energy, used for growing and maintenance, and FM represents the fasting maintenance losses, which are usually assumed to depend on body weight and temperature:

$$\text{FM}(BW, T) = a \times BW^b \times e^{c \times T}$$

where BW is the body weight, T is temperature and a, b and c are estimated parameters. The “b” parameter is often called the “body weight exponent” and has biological significance as it represents how costs scale with fish size.

Bioenergetic often have significant limitations, in describing the chemical composition (i.e., moisture, protein, lipid, and ash) of biomass gain (Dumas et al., 2010). Additionally, they can present several drawbacks when applied under the overly simplistic assumption that the energy density of fish is constant (Canale and Breck, 2013; Dumas et al., 2010). In turn, nutrient-based models, such as energy-protein flux models (EP models), besides energy, also consider protein intake to predict growth and body composition based on protein retention equations along with fish weight and temperature (Nobre et al., 2019).

In these models, maintenance requirements are influenced by body size and temperature, whereas growth requirements are influenced by the magnitude and composition of weight gain (D. P. Bureau et al., 2003; Lupatsch et al., 1998, 2003b, 2003a; Lupatsch and Kissil, 2005). Thus, the metabolic expenditure (protein and energy losses) for maintenance can be characterized by an exponent that determines the change in metabolic rate as a function of body weight (Lupatsch and Kissil, 2005). Thus, in this work, we consider an alternative parametrization of the model mentioned above, where an explicit “retention efficiency coefficient” (k) replaces the COG parameter, applying it both to energy and protein retention:

$$\text{RE} = k_E \times \text{DE} - \text{FM}_E$$

$$\text{RP} = k_P \times \text{DP} - \text{FM}_P$$

The estimation of these exponents is already documented for different fish families (Clarke and Johnston, 1999) and species, such as barramundi *Lates calcarifer* (B. D. Glencross, 2008), gilthead seabream *Sparus aurata* (Lupatsch et al., 1998), European seabass *Dicentrarchus labrax*, white grouper *Epinephelus aeneus* (Lupatsch et al., 2003a; Lupatsch et al., 2001a) and yellowtail kingfish *Seriola lalandi* (Booth et al., 2010), and generally similar metabolic weight exponents of energy and protein have been reported. Therefore, it is broadly accepted to assume standard metabolic body weight exponents of 0.80 and 0.70 for energy and protein, respectively. Though it is possible to estimate the metabolic body weight exponents for energy and protein using calibration data related to experiments performed either with fed and unfed fish, it is not clear whether it is preferable or more correct to do so over assuming fixed standard exponents.

There are still some scientific and technical questions regarding the implementation of nutrient-based models, which perhaps hampers their more widespread adoption in commercial settings. One could be the lack of transparency regarding the way models are developed and the assumptions that are made during calibration, which often makes them impossible or very difficult to replicate. During the model development process, it is important to assess both calibration and validation errors. First, one should calibrate models under different assumptions (i.e., using different calibration methods) and estimate the performance of the calibration process (e.g., through cross-validation) in generating models that effectively predict new data. This evaluation should be carried out as objectively as possible, considering not only quantitative criteria (e.g., error metrics), but also qualitative ones (e.g., diagnostic plots), in order to consistently ensure an acceptable predictive ability of the models. Ultimately, to ensure that the model can effectively predict new data (i.e., data not used during the calibration process), and to fairly quantify expected prediction errors, it is necessary to perform a validation with an independent data set and assess validation errors.

There is a need to explore and develop models that can be fairly easy to use by fish farmers and still have the ability to predict fish growth and composition based on fundamental properties of fish metabolism. Therefore, there must be an effort by the scientific community to overcome some scientific and technical issues regarding the implementation of these models by the industry and thus bridge the gap between both.

Herein, Nile tilapia was chosen as a model fish due to its aquaculture importance, since it is one of the most economically important fish species in aquaculture. Moreover, it is a widespread distribution species - which in part is linked to its remarkable ability to tolerate diverse environmental condition (e.g., water temperatures, salinities and oxygen levels). Additionally, it is omnivorous species and thus can thrive on a variety of diets. These makes Nile tilapia an ideal species for investigating responses not only to environmental, but also to different nutritional stressors. In this work, Nile tilapia growth data were collected and used to compare different combinations of models and calibration methods, using objective criteria based on cross-validation, to determine the best combination for this species. Additionally, we also sought to clarify the relative merit of different biological and technical hypotheses

by calibrating models under these different assumptions and comparing them in terms of their capacity to generalize for new data.

2. Materials and methods

2.1. Data collection and analysis

Nile tilapia growth data were collected from the literature to be further used for model calibration and validation, covering a wide range of rearing conditions and feed composition (Table 1 and Table 2, respectively). All datasets were converted into a standard format where information on the growth, body composition, feed conversion ratio (FCR), water temperature and diet properties are stored on a daily resolution basis. Missing data was handled by using default values (e.g., apparent digestibility coefficients) or by applying interpolation methods (e.g., daily feed intake was estimated based on FCR and growth, when not reported in literature).

An exploratory analysis of calibration data was performed to see how relevant responses were affected by different factors (e.g., body weight, feed intake, digestible protein intake, digestible energy intake) and to obtain information that could be relevant for model development. This analysis consisted mainly of scatter plot visualization and trend line evaluation (linear vs. non-linear) of ingested against retained protein and energy fractions, using the F-test to assess the statistical significance of results.

2.2. Model evaluation and calibration methods

Growth models with diverse levels of complexity (FCR, bioenergetic, energy-protein flux) were developed. In addition, 2 models already published by other authors (Chowdhury et al., 2013; van Trung et al., 2011) were recalibrated with our dataset in order to fairly evaluate their performance. Different calibration methods were implemented: by considering either least squares or Huber loss; by considering either fixed standard or estimated body weight exponents for the fasting maintenance and body composition sub-models; and by considering either only fixed effects or both fixed and random (mixed model) effects during model calibration. Body weight exponents were estimated by considering that protein/energy loss in fasted fish is proportional to body weight (BW) raised to a certain exponent. Thus, by applying multilinear regressions of log(loss) as a function of log(BW) and temperature, the BW exponents are recovered as one of the coefficients. To complement the limited data available for Nile tilapia under fasting conditions, the exponent estimates also considered (with low weight) fasting loss estimates obtained for growing fish by extrapolation of the trial-wise gain/intake relationships to 'zero intake'. Therefore, by calibrating the different models under the different sets of assumptions (see Fig. 1), 73 combinations of "models + calibration methods" were obtained and their performance was evaluated. The combinations of "model + calibration method" were evaluated qualitatively by visual examination of model behaviour (e.g., observed vs. predicted plots) and residuals (e.g., Q-Q plots), and quantitatively by calculating the mean

Table 1
Summary of collected data used to calibrate models.

Attributes	Unit	Nile tilapia
Nr. of data sources	–	27
Nr. of diets	–	150
Body weight range	g	0.51–457
Temperature range	°C	23–29
FCR	–	0.9–3.9
Diet composition range		
Crude protein	% as fed	23–46
Crude fat	% as fed	4–15
Gross energy	MJ/kg	14–20

Table 2

Summary of the independent dataset used to validate models.

Attributes	Unit	Nile tilapia
Nr. of data sources	–	16
Nr. of diets	–	103
Body weight range	g	0.51–571
Temperature range	°C	21.4–30
FCR	–	0.9–3.5
Diet composition range		
Crude protein	% as fed	22–45
Crude fat	% as fed	2–15
Gross energy	MJ/kg	13–21

absolute percentage error (MAPE) and root mean square error (RMSE), as follows:

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100$$

$$RMSE(g) = \sqrt{\frac{\sum_{i=1}^n (\hat{y}_i - y_i)^2}{n}}$$

where, $\hat{y}_i$ is the predicted value, y_i is the observed value and n the number of predicted-observed value pairs.

Moreover, in addition to calibration error metrics, focus was given on cross-validation errors obtained through different cross-validation methods (k-fold cross-validation with $k = 10$, $k = 5$ and $k = 2$, with 5, 10 and 25 repetitions, respectively), to ensure that the chosen models display the best performance when predicting outside their calibration set. To determine the risk of overfitting and to decrease the degree of subjectivity, cross-validation was performed with different k values in order to subject the combinations to different levels of "difficulty", since, as the k value decreases, less data is available to calibrate the model.

All analyses were performed using R version 4.1.2 (R Core Team, 2021), where the *rlm* function from the MASS package (Venables and Ripley, 2002) was used to calibrate the robust models and *nlme* function from the nlme package (Pinheiro and Bates, 2000) was used to calibrate mixed-effects models.

2.3. Validation of calibrated models

From the 73 "model + calibration methods" combinations, a representative sample of 12 models of different types and with different levels of performance were chosen to be validated. In addition to the models from Chowdhury et al. (2013) and van Trung et al. (2011) that were recalibrated with our data, the same models were also validated using their original parametrization (BE_Chow_par model from Chowdhury et al., 2013; EPG1_par model from van Trung et al., 2011).

3. Results

3.1. Data analysis

Exploratory data analysis suggests a direct relationship between digestible energy intake and energy gain, and between digestible protein intake and protein gain (Fig. 2). It appears that there is better efficiency in protein gain, since the data tends to be closer to the $y = x$ line than the energy data. Moreover, since both linear and quadratic models fit the data accurately, there is no strong evidence for a saturation effect at higher intake levels, on either energy or protein gain. On the other hand, though digestible protein intake does not appear to affect protein retention efficiency (F-test $P > 0.05$), which is consistent with the lack of a saturation effect, the same was not observed for digestible energy intake, where a negative effect on energy retention efficiency (F-test, $P < 0.001$) was observed (Fig. 3).

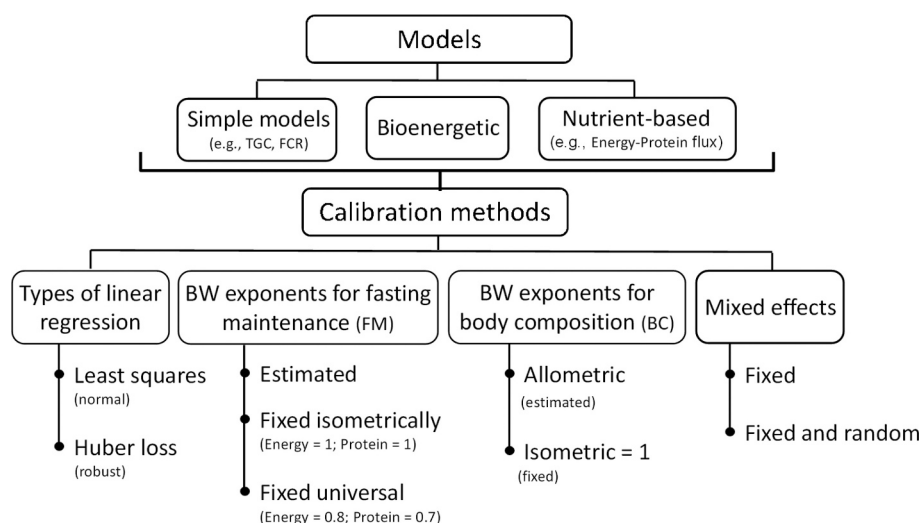


Fig. 1. Diagram explaining the different types of models and calibration methods that have been developed based on different assumptions.

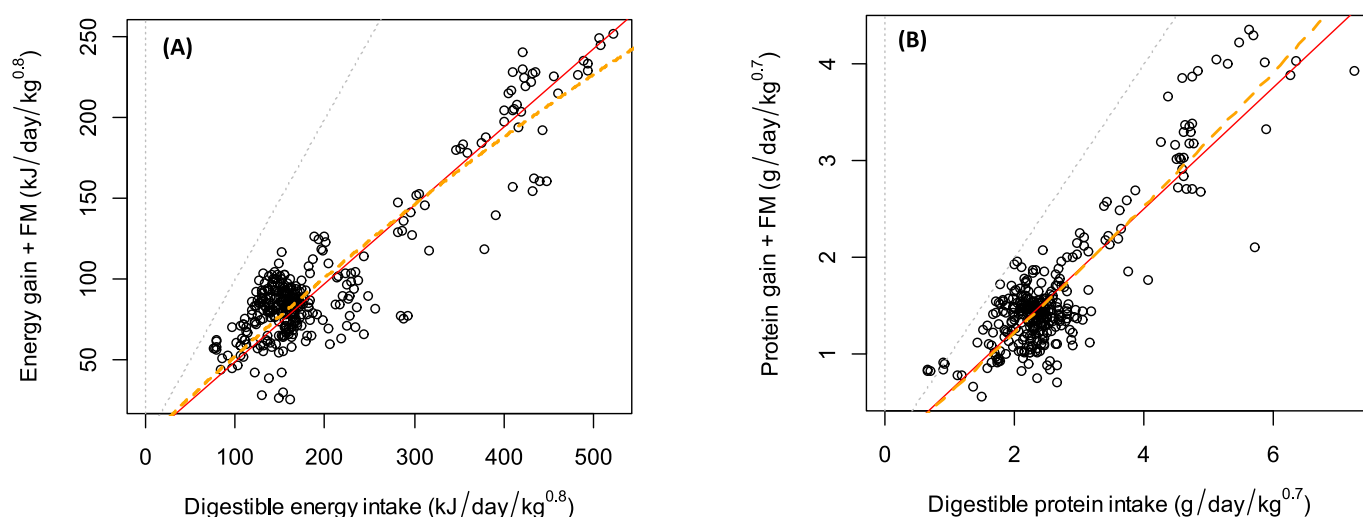


Fig. 2. Scatter plot showing the relation between digestible energy intake ($\text{kJ/day/kg}^{0.8}$) and energy gain ($\text{kJ/day/kg}^{0.8}$) (A), and the relation between digestible protein intake ($\text{g/day/kg}^{0.7}$) and protein gain ($\text{g/day/kg}^{0.7}$) (B), corrected by adding the fasting maintenance (FM) costs (to ensure a zero intercept). Black points represent measured values. Red line represents linear regression. Dashed orange line represents quadratic regression. Grey dotted line represents the “ $y = x$ ” line (theoretical maximal response). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Regarding the effect of body weight, while there is no evidence of an effect on protein retention efficiency (F-test $P > 0.05$), a positive effect of fish body weight on energy retention efficiency (F-test, $P < 0.001$) was observed (Fig. 4). Furthermore, no effect was found for DP/DE ratio on energy retention efficiency (F-test, $P > 0.05$) (Fig. 5 A). However, it seems that the DP/DE ratio has a negative effect on protein retention efficiency (F-test, $P < 0.001$) (Fig. 5 B).

3.2. Model calibration and evaluation

Considering the results of cross-validation, nutrient-based models, like the EP model (energy-protein flux), generally appear to provide better predictions than the simpler bioenergetic models. Overall, the results of the different cross-validations were consistent, in the sense that the best and worst performing models were generally the same regardless of the type of cross-validation used. However, it is important to mention that the overall agreement was not entirely exact, since each cross-validation method presented a slightly different order/classification for models (Table 3).

In particular, EP models displayed calibration and cross-validation

errors between 7 and 11% for body weight, and between 12 and 14% for body composition. In turn, bioenergetic models had calibration and cross-validation errors between 9 and 26% for body weight, and $\approx 13\%$ for body composition predictions (Table 3). Additionally, models had better prediction capabilities when calibrated under the assumption of fixed standard body weight exponents for fasting maintenance (0.80 and 0.70 for energy and protein, respectively) or with isometric ones (exponents = 1). By visually analysing the models' behaviour, it was evidenced that the same model can display distinct behaviours when calibrated with different methods (see, for example, Fig. 6).

The model's complexity seemed to affect its predictive capability (Table 3). Overall, models that only considered energy (i.e., BE models) to predict growth had higher prediction errors than models that also consider protein besides energy (i.e., EP models). Furthermore, bioenergetic models, like BE, displayed poor growth predictions and additionally presented large discrepancies in the initial predictions of body composition (Fig. 6). This was further supported by analysis of the calculated metrics (e.g., MAPE), in which cross-validation errors were higher than calibration ones. The bioenergetic model obtained from Chowdhury et al. (2013) (BE_{Chow}) was the one with the highest

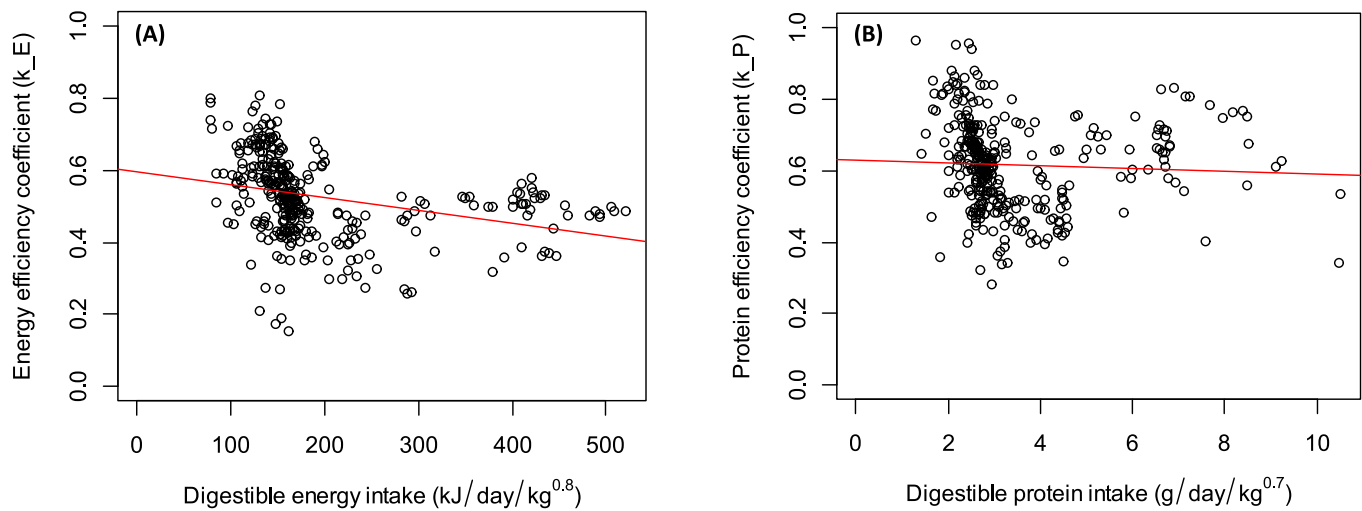


Fig. 3. Scatter plot showing the effect of digestible energy intake ($\text{kJ/day/kg}^{0.8}$) on energy retention efficiency (A) and the effect of digestible protein intake ($\text{g/day/kg}^{0.7}$) on protein retention efficiency (B). Black points represent measured values. Red line represents linear regression. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

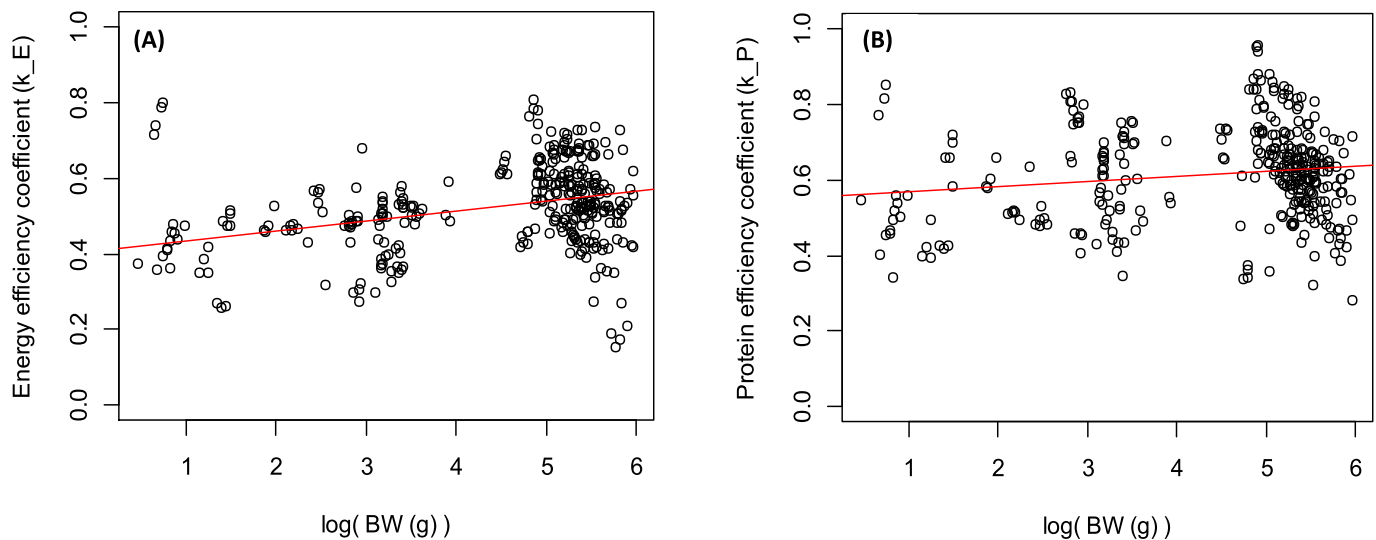


Fig. 4. Scatter plot showing the effect of fish body weight (g) on energy (A) and protein (B) retention efficiency. Black points represent measured values. Red line represents linear regression. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

calibration and cross-validation errors, with a *MAPE* of 26% for body weight and $\approx 13\%$ for body composition. In turn, the *BECOG_iso* model was the bioenergetic model with the best performance, showing calibration and cross-validation errors between 9 and 12% for body weight, and $\approx 14\%$ for body composition.

The *van Trung et al. (2011)* model (*EPG1*) performed better than the *BE_Chow*, obtaining calibration errors with a *MAPE* 9% for body weight and $\approx 15\%$ for body composition; cross-validation errors with a *MAPE* of 12% for body weight and $\approx 16\%$ for body composition was also obtained (Table 3). Interestingly, the *FCR* model had better predictions than the bioenergetic models in general and the *EPG1* model, in general, showing calibration errors with a *MAPE* of $\approx 11\%$ for body weight and $\approx 13\%$ for body composition, and cross-validation errors with a *MAPE* $\approx 10\%$ for body weight and $\approx 12\%$ for body composition (Table 3).

The metabolic body weight exponents were estimated for the calibration dataset (Table 1) by different methods and varied between 0.71 and 0.77 for energy and 0.76–0.78 for protein. However, models in which the standard metabolic body weight exponents (0.80 and 0.70, for energy and protein, respectively) were assumed showed a better

predictive capability than methods that considered the estimated exponents (Table 3).

3.3. Model validation

Out of all tested models, the *EP5_mixed* (energy-protein model calibrated assuming mixed effects) and the *EP5_u_rlm* models (energy-protein model calibrated with Huber loss linear regression and with fixed body weight exponents for maintenance costs) displayed the best performance: both models presented similar validation errors, with a *MAPE* of 9% for body weight (Table 4). In qualitative terms, both models predicted the validation data well, demonstrating a reasonably capacity to predict Nile tilapia growth and composition (Fig. 6). Although both *EP5_mixed* and *EP5_u_rlm* displayed similar validation errors in predicting body weight, the *EP5_mixed* presented slightly lower validation errors when predicting body composition (*MAPE* of 7% vs. 8%), which may indicate more accurate body composition predictions (See Figs. 7 and 8).

Overall, the bioenergetic models have worse performance in

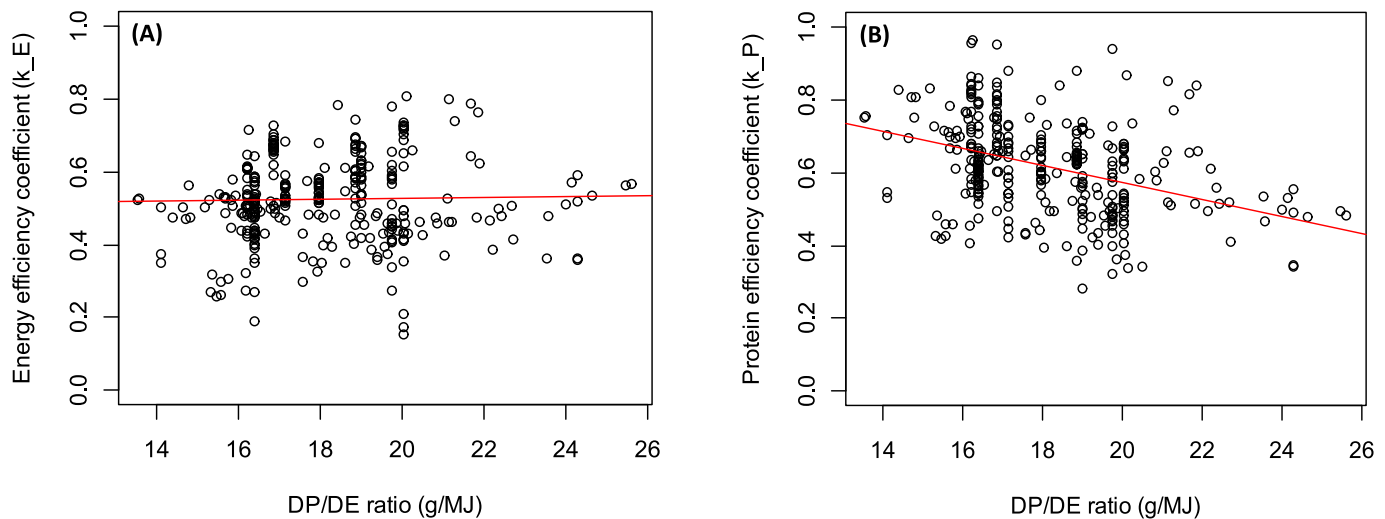


Fig. 5. Scatter plot showing the effect of digestible protein and digestible energy ratio (DP/DE) (g/MJ) on energy (A) and protein (B) retention efficiency. Black points represent measured values. Dashed orange line represents linear regression.

predicting Nile tilapia growth than *EP* models, though *BECOG_{iso}* seems to be the best tested bioenergetic model ($MAPE_{bw} \approx 13\%$ and $MAPE_{bc} \approx 8\%$). Nevertheless, the tested models show fair predictions for Nile tilapia growth, except for the *BE_{Chow}* ($MAPE_{bw} = 28\%$ and

$MAPE_{bc} = 7\%$), *EPG1_{par}* ($MAPE_{bw} = 34\%$ and $MAPE_{bc} = 116\%$) and *BE_{Chow_{par}}* ($MAPE_{bw} = 178\%$ and $MAPE_{bc} = 10\%$).

The FCR model displayed the lowest RMSE validation error (16.81 g) of all tested models and one of the lowest MAPE values for both weight and body composition ($MAPE_{bw} = 11\%$ and a $MAPE_{bc} = 7\%$).

The Chowdhury et al. (2013) and van Trung et al. (2011) models seem to perform better when parameterised with the dataset used in this study

(*BE_{Chow}* $MAPE_{bw} = 27\%$ vs. *BE_{Chow_{par}}* $MAPE_{bw} = 178\%$;

EPG1 $MAPE_{bw} = 10\%$ vs. *EPG1_{par}* $MAPE_{bw} = 34\%$). However, regardless of the parameterization used, the model retrieved from Chowdhury et al. (2013) performed better when body composition is updated based on the composition of the body weight gained (*BE_{Chow2}* and *BE_{Chow2_{par}}*), rather than considering the whole-body weight of the fish (*BE_{Chow}* and *BE_{Chow_{par}}*) (Table 3).

4. Discussion

Mathematical models can be useful tools to describe the various processes involved in fish growth. It is necessary to understand the processes to be modelled, and careful and thorough data collection and analysis is of utmost importance. This will help describe and clarify the relationships inherent to the processes, before starting developing the model. In this context, the relationship between protein and energy intake, and deposition is one of the key aspects underlying fish growth. The data collected in this work pointed towards a linear relationship between digestible energy intake and energy gain for Nile tilapia. This is consistent with what has been previously demonstrated by Lupatsch et al. (2010) for Nile tilapia and by Lupatsch et al. (2001a, b, 2003a, b) for gilthead seabream, but does not support the quadratic relationship model that van Trung et al. (2011) reported for Nile tilapia. Likewise, the relationship between digestible protein intake and protein gain is linear, confirming previous observations (Lupatsch et al., 1998, 2010). Nevertheless, other studies reported a quadratic relationship in Nile tilapia (van Trung et al., 2011) and other species (Booth et al., 2010; B. Glencross et al., 2011; B. D. Glencross, 2008; Lupatsch et al., 2001b). Konnert et al. (2022) have recently stated that protein gain in Nile tilapia follows a two-phased response to increasing protein intake, being described by a linear-plateau model. Their model indicates that protein

deposition is linear until $8.4 \text{ g kg}^{0.8} \text{ day}^{-1}$ and then levels-off. In the present study instead of a level-off pattern, a linear relationship was observed, possibly because of data scarcity above $8 \text{ g kg}^{0.8} \text{ day}^{-1}$. This is mainly because most of the trials in the literature do not overfeed tilapia with fat or protein, making it impossible to obtain data showing the effect of saturation on energy or protein gain. Thus, what the results show is that the linear model works better than the quadratic model to show the relationship between intake and gain when Nile tilapia is fed diets that are not extremely dense in terms of protein or fat composition. Furthermore, our analysis shows that the energy retention efficiency is body weight-dependent. Likewise, Breck (2011), showed that fish energy density varies with body size and should be considered to enhance the performance of bioenergetic models. Moreover, DP/DE was shown to negatively affect protein retention efficiency, supporting previous studies by Haidar et al. (2018) and Konnert et al. (2022) in Nile tilapia. But this does not apply to energy retention efficiency, as Konnert et al. (2022) had already reported for Nile tilapia.

During model evaluation, the quantitative and qualitative analyses were combined with different cross-validation methods to objectively identify the best models and calibration methods. Though generally consistent, the results of these were not always entirely in agreement due to how each method penalizes models according to their complexity: methods with lower k -values are more restrictive and therefore exert a higher penalty on more complex models, than methods with higher k -values (Hastie et al., 2001).

After the calibration and cross-validation process, the models were validated with an independent Nile tilapia dataset, where *EP5_{u_{rlm}}* and *EP5_{mixed}* displayed the lowest validation errors and thus, were considered the best models to be used to predict Nile tilapia growth. However, it should be noted that the *EP5_{mixed}* model has slightly lower validation errors for predicting body composition than *EP5_{u_{rlm}}*. This difference may be associated with the fact that the *EP5_{mixed}* model uses a calibration method where body composition is predicted with mixed-effect models. One advantage of using this type of approach is that it allows to overcome problems related to data heterogeneity (Schielzeth et al., 2020). The dataset used in this study is composed by data collected from different sources with distinct number of data/points, and each one with different conditions (e.g., types of culture, feeds, temperature, strains) that somehow can affect fish growth and body composition. Moreover, mixed-effect models are more flexible and thus overcome violations, when present, in distributional assumptions (Gelman and Hill, 2006; Schielzeth et al., 2020). Furthermore, such models, can determine the appropriate shrinkage (i.e., partial pooling) for low-

Table 3

Representative sample of the developed models.

2-Fold CV	Model acronym	CALIBRATION			CROSS-VALIDATION		
		MAPE _{bw} (%)	RMSE (g)	MAPE _{bc} (%)	MAPE _{bw} (%)	RMSE (g)	MAPE _{bc} (%)
Energy-protein linear regression, BWexp = fixed	EP5_u_iso	7,53	10,59	12,27	10,49	16,39	13,96
Energy-protein linear regression, mixed effects to calculate all parameters	EP5_mixed	8,00	10,49	12,13	10,94	16,41	13,15
Energy-protein robust regression, BWexp = fixed	EP5_u_rlm_iso	7,47	10,58	13,09	10,08	15,89	14,99
Hybrid Energy-protein allometric/isometric ensemble average, linear regression, FM BWexp = fixed, BC BWexp = estimated	EP5hyb_u	7,55	10,20	12,43	10,56	16,26	14,20
Energy-protein linear regression, FM BWexp = fixed	EP5_u	7,63	10,04	12,54	10,65	16,20	14,33
Energy-protein linear regression, BC BWexp = fixed	EP5_iso	7,30	9,92	12,23	11,43	16,89	13,72
Energy-protein robust regression, FM BWexp = fixed	EP5_u_rlm	7,58	10,11	13,48	10,23	15,64	15,49
Energy-protein linear regression, mixed effects only to calculate FM	EP5_mixed_fm	8,63	10,78	12,66	10,84	15,50	13,42
FCR , BWexp = estimated	FCR	9,77	16,57	13,40	11,32	17,00	13,41
Bioenergetic model , linear regression FM BWexp = estimated, BC BWexp = fixed Fish energy density body weight-dependent, COG body weight-dependent	BECOG_iso	9,19	21,39	13,85	11,81	33,89	13,96
*Energy-protein quadratic , BWexp = estimated	EPG1	8,55	10,44	14,82	11,97	15,57	19,70
**Bioenergetic , linear regression, FM BWexp = estimated, BC BWexp = fixed, Fish energy density constant. Body composition is updated based on the composition of body weight gain.	BE_Chow2	13,53	36,94	12,73	14,64	37,40	12,72
Bioenergetic model , linear regression BWexp = estimated, Fish energy density body weight-dependent, COG constant	BE	16,25	22,14	13,39	18,55	36,62	13,42
**Bioenergetic , linear regression, FM BWexp = estimated, BC BWexp = fixed, Fish energy density constant. Body composition is update based on whole-body weight	BE_Chow	25,68	57,47	13,11	25,89	57,68	13,10
5-Fold CV							
Energy-protein robust regression, BWexp = fixed	EP5_u_rlm_iso	7,81	10,92	12,46	9,58	11,10	12,98
Hybrid Energy-protein allometric/isometric ensemble average, linear regression, FM BWexp = fixed, BC BWexp = estimated	EP5hyb_u	8,18	10,89	12,00	10,38	11,53	12,58
Energy-protein linear regression, BWexp = fixed	EP5_u_iso	8,13	11,55	11,82	10,31	11,86	12,36
Hybrid Energy-protein allometric/isometric ensemble average, robust regression, FM BWexp = fixed, BC BWexp = estimated	EP5hyb_u_rlm	7,87	10,38	12,63	9,64	10,76	13,17
Energy-protein linear regression, BWexp = isometric	EP5_iiso	9,02	11,92	11,78	10,67	12,29	12,39
Energy-protein linear regression, FM BWexp = fixed	EP5_u	8,26	10,54	12,12	10,48	11,37	12,74
Energy-protein robust regression, FM BWexp = fixed	EP5_u_rlm	7,96	10,01	12,90	9,72	10,52	13,45
Energy-protein linear regression, mixed effects for all parameters	EP5_mixed	8,65	12,91	11,95	10,61	12,09	12,41
FCR , BWexp = estimated	FCR	10,49	16,76	12,96	10,05	13,81	12,39

Bioenergetic model , linear regression FM BWexp = estimated, BC BWexp = fixed Fish energy density body weight-dependent, COG body weight-dependent	<i>BECOG_iso</i>	8,93	18,79	13,33	10,27	17,35	12,68
**Bioenergetic , linear regression, FM BWexp = estimated, BC BWexp = fixed, Fish energy density constant. Body composition is updated based on the composition of body weight gain.	<i>BE_Chow2</i>	14,49	39,80	12,30	14,04	32,07	11,63
Bioenergetic model , linear regression BWexp = estimated, Fish energy density body weight-dependent, COG constant	<i>BE</i>	17,86	19,85	13,02	20,40	23,27	12,34
*Energy-protein quadratic , BWexp = estimated	<i>EPG1</i>	9,01	10,59	14,86	11,46	12,24	15,76
**Bioenergetic , robust regression, BWexp = estimated, BC BWexp = fixed. Fish energy density constant. Body composition is update based on whole-body weight	<i>BE_Chow</i>	26,30	58,74	12,70	25,86	48,83	12,12

10-Fold CV

Energy-protein linear regression, BWexp = isometric	<i>EP5_iiiso</i>	9,12	11,43	11,78	9,90	9,53	12,93
Energy-protein linear regression, mixed effects for all parameters	<i>EP5_mixed</i>	8,81	12,87	12,06	9,77	9,80	13,31
Hybrid Energy-protein allometric/isometric ensemble average, robust regression, FM BWexp = fixed, BC BWexp = estimated	<i>EP5hyb_u_rlm</i>	8,06	10,42	12,61	9,26	9,25	14,40
Energy-protein robust regression, FM BWexp = fixed	<i>EP5_u_rlm</i>	8,16	10,04	12,88	9,31	9,03	14,70
Energy-protein robust regression, BWexp = fixed	<i>EP5_u_rlm_iso</i>	7,99	10,98	12,44	9,22	9,56	14,20
Energy-protein linear regression, BWexp = fixed	<i>EP5_u_iso</i>	8,36	11,55	11,83	9,88	9,99	13,41
Hybrid Energy-protein allometric/isometric ensemble average, linear regression, FM BWexp = fixed, BC BWexp = estimated	<i>EP5hyb_u</i>	8,42	10,87	12,01	9,93	9,76	13,63
FCR , BWexp = estimated	<i>FCR</i>	10,55	16,88	13,02	9,88	12,26	12,24
Bioenergetic model , linear regression FM BWexp = estimated, BC BWexp = fixed, Fish energy density body weight-dependent, COG body weight-dependent	<i>BECOG_iso</i>	8,97	16,91	13,37	10,17	14,28	12,63
**Bioenergetic , linear regression, FM BWexp = estimated, BC BWexp = fixed, Fish energy density constant. Body composition is updated based on the composition of body weight gain.	<i>BE_Chow2</i>	14,70	41,42	12,37	14,12	28,58	11,45
Bioenergetic model , linear regression BWexp = estimated, Fish energy density body weight-dependent, COG constant	<i>BE</i>	20,87	18,86	13,11	23,26	19,70	12,29
*Energy-protein quadratic , BWexp = estimated	<i>EPG1</i>	9,34	10,72	14,97	10,98	10,33	15,89
Energy-protein linear regression, FM BWexp = fixed	<i>EP5_u</i>	9,24	14,08	12,94	10,35	12,50	14,52
**Bioenergetic , robust regression, FM BWexp = estimated, BC BWexp = fixed. Fish energy density constant. Body composition is update based on whole-body weight	<i>BE_Chow</i>	26,43	59,92	12,75	25,96	43,75	11,94

Model's error metrics calculated for each method of cross-validation. The greener cells indicate lower estimates of the generalization errors, and the reddish ones indicate higher ones. MAPE_{bw} = Mean absolute percentage error for body weight; MAPE_{bc} = Mean absolute percentage error for body composition. FM = fasting maintenance; BC = body composition; BWexp = body weight exponents. COG = costs of growth.

*van Trung et al., 2011.

**Chowdhury et al., 2013

sample groups, pushing extreme values towards the mean (Schielzeth et al., 2020) and therefore, no matter how many points each source has, all will have balanced weights when performing regression analysis. The use of mixed effects models is therefore strongly advised when using heterogeneous datasets for model calibration.

In this study, different assumptions were made when developing bioenergetic models, which were based on the preliminary data analysis. For instance, for the *BE* model it was assumed that the fish energy density was body weight-dependent, but costs of growth (COG) were

constant, whereas in *BECOG* both the costs of growth and fish energy density were body weight-dependent. The results showed that assuming that fish energy density and cost of growth are both body weight-dependent seems to be the most correct assumption, as the *BECOG* model proved to be the best performing bioenergetic model. Moreover, these results emphasize that the energy retention efficiency of Nile tilapia is seemingly body weight-dependent and should be taken into consideration when modelling growth (Breck, 2011), as also suggested by the preliminary data analysis.

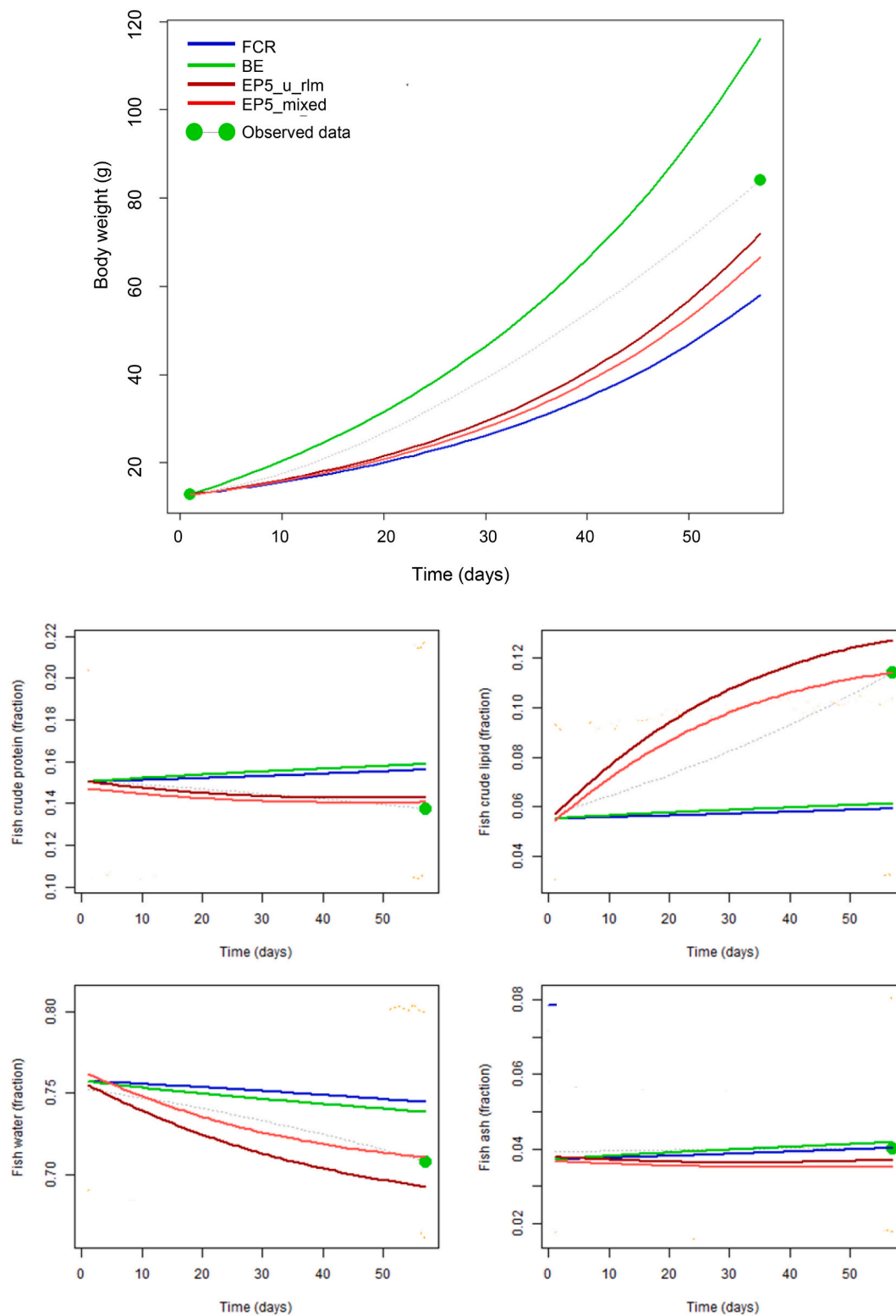


Fig. 6. Line plots showing predicted growth and body composition along time for different models, taking one of the calibration/validation datasets as prediction target. Green points represent measured values. Lines represent either an interpolated body composition value (dashed grey line) or a model prediction (full coloured lines). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The models that were developed in this study presented lower validation errors when compared to those previously reported by [van Trung et al. \(2011\)](#) and

[Chowdhury et al. \(2013\)](#) for Nile tilapia. In fact, when these models are parameterized with the dataset used in this study (*BE_Chow* and

EPG1), they display better performance than when using the original parametrization obtained by the authors (*BE_Chow_par* and *EPG1_par*), highlighting the relevance of the dataset used to calibrate fish growth models as they can affect their prediction quality. Other factors may explain the poor performance of *BE_Chow* and *BE_Chow_par* models, such

Table 4

Validation errors metrics for each model after validation with an independent dataset.

Model	Model acronym	MAPE _{bw} (%)	RMSE (g)	MAPE _{bc} (%)
Energy-protein robust regression, FM BWexp = fixed	<i>EP5_u_rlm</i>	9,65	31,64	7,71
Hybrid Energy-protein allometric/isometric ensemble average, robust regression, FM BWexp = fixed, BC BWexp = estimated	<i>EP5hyb_u</i>	9,74	33,40	7,71
Energy-protein linear regression, mixed effects for all parameters	<i>EP5_mixed</i>	9,87	32,49	6,81
Energy-protein robust regression, BWexp = fixed	<i>EP5_u_rlm_iso</i>	9,88	35,41	8,01
Energy-protein linear regression, FM BWexp = fixed	<i>EP5_u</i>	10,00	33,67	7,26
*Energy-protein quadratic , BWexp = estimated	<i>EPG1</i>	10,13	32,13	9,13
Energy-protein linear regression, BWexp = fixed	<i>EP5_u_iso</i>	10,19	37,24	7,63
FCR , BWexp = estimated	<i>FCR</i>	11,40	16,81	6,47
Bioenergetic model , linear regression FM BWexp = estimated, BC BWexp = fixed Fish energy density body weight-dependent, COG body weight-dependent	<i>BECOG_iso</i>	12,74	49,11	7,98
**Bioenergetic , robust regression, FM BWexp = estimated, BC BWexp = fixed, Fish energy density body weight-dependent, COG constant	<i>BE_Chow2</i>	16,60	57,20	6,67
Bioenergetic model , linear regression BWexp = estimated, Fish energy density body weight-dependent, COG constant	<i>BE</i>	19,37	22,46	6,37
**Bioenergetic , robust regression, Fish energy density body weight-dependent, COG constant, with Chowdhury et al. (2013) parameterisation	<i>BE_Chow2_par</i>	21,18	57,14	9,51
**Bioenergetic , robust regression, FM BWexp = estimated, BC BWexp = fixed	<i>BE_Chow</i>	27,38	81,77	6,47
*Energy-protein quadratic with van Trung et al. (2011) parameterisation	<i>EPG1_par</i>	33,73	59,68	115,78
**Bioenergetic , robust regression, with Chowdhury et al. (2013) parameterisation	<i>BE_Chow_par</i>	178,07	917,40	9,73

The greener cells indicate lower estimates of the generalization errors, and the reddish ones indicate higher ones. MAPE_{bw} = Mean absolute percentage error for body weight; MAPE_{bc} = Mean absolute percentage error for body composition. FM = fasting maintenance; BC = body composition; BWexp = body weight exponents. COG = costs of growth.

*van Trung et al., 2011.

** Chowdhury et al., 2013

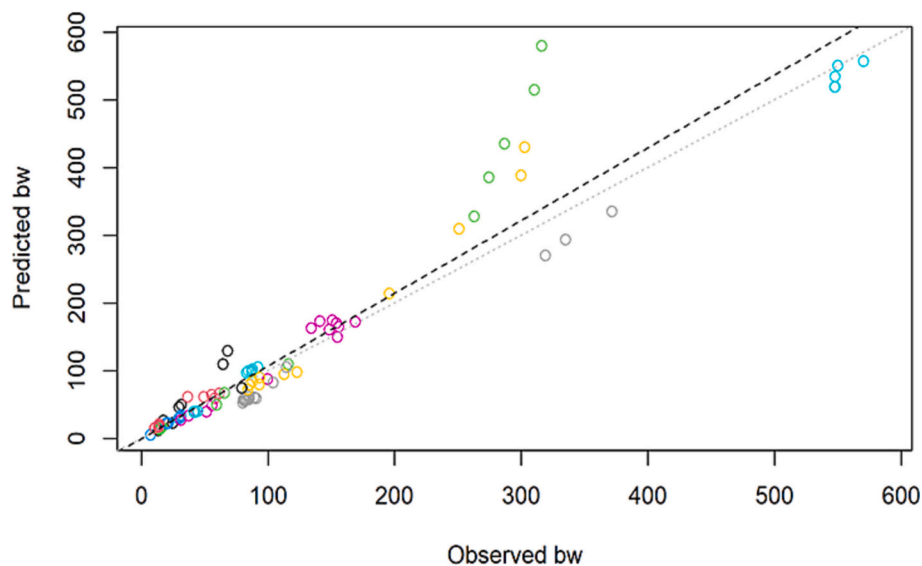


Fig. 7. Scatter plot showing the differences between the observed and the predicted body weight points of EP5_fast_u_rlm model in log scale. Segmented black line represents the linear regression of y as a function of x and the dotted grey line represents the y = x line.

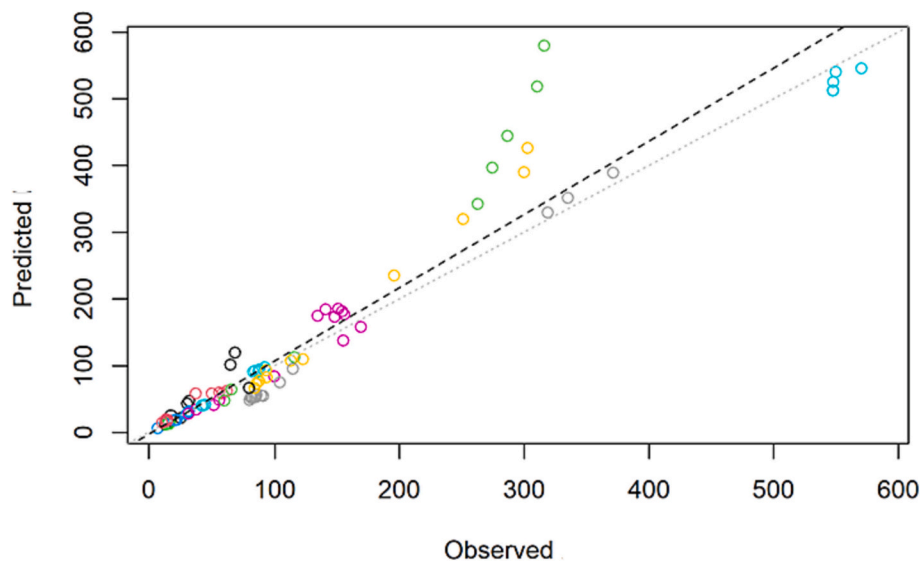


Fig. 8. Scatter plot showing the differences between the observed and the predicted body weight points of EP5_fast_mixed model in log scale. Segmented black line represents the linear regression of y as a function of x and the dotted grey line represents the $y = x$ line.

as the fact that protein intake is not considered and that the update of body composition is forced to follow the body composition model in Chowdhury et al. (2013).

Another important factor that affects models' performance is how the different body components (protein, lipids, water and ash) are updated and predicted. In this work, two different approaches were considered: a) predict body composition with a static model, directly as a function of fish body weight (e.g., *BE*, *FCR*, *BE_Chow* models); or b) use a dynamic model that updates water and ash according to protein retention, which in turn is calculated based on feed intake and feed composition. The first approach displayed large discrepancies in the initial predictions of body composition when compared to the observed ones, besides the poor predictions for growth. In turn, when the body composition is predicted based on protein retention equations (e.g., *EP* models), models displayed accurate body composition and growth predictions. However, the *EP* models when considering energy do not include the distinction between fat and carbohydrates. Therefore, this type of models may perform better for carnivorous species because the energy input comes mainly from fat, while omnivorous species such as Nile tilapia use fat, protein and carbohydrates as energy sources (Boonuntanasarn et al., 2018; Stone, 2010). Interestingly, the *FCR* model, although one of the simplest tested models, displayed better performance than the bioenergetic ones. Although *FCR* had shown lower absolute errors (RMSE), they translate into higher relative errors for small fish, which explains the suboptimal MAPE displayed by this method. The overall good performance displayed by this method can be explained, at least in part, by the fact that the daily feed intake in the dataset used in this study was often estimated based on *FCR* values reported in each source, which may provide the *FCR* model an unfair advantage. In addition, the good performance of *FCR* may also be related to tilapia dietary requirements when compared to other finfish species. Nile tilapia can digest carbohydrates well (Maas et al., 2020; NRC, 1993) and thus grow well when fed carbohydrate-rich and protein-poor diets (Bomfim et al., 2008; Botaro et al., 2007; Furuya et al., 2005). Therefore, since *FCR* ignores variations in diet properties, it makes the model perform somewhat reasonably well, even under extreme variations of these factors. However, the *EP* models had $\approx 3\%$ more prediction capability compared to *FCR*, which shows that it is however still important to take protein input into account when modelling tilapia growth.

Furthermore, the type of relationship between energy/protein intake and gain is also important to be considered. In fact, this is the main

difference between the van Trung et al. (2011) models (*EPG1* and *EPG1_par*) and the *EP* models developed in this study: the *EPG1* uses quadratic relationships, while the *EP5* model assumes linear relationships. This does not mean that the van Trung et al. (2011) approach was necessarily incorrect, but rather that the assumptions made in model development are highly dependent on dataset selection. As previously mentioned, the data collected in this study do not describe any kind of quadratic but linear relationship between energy and protein. Therefore, linear equations were considered when developing the models. It is also for this reason that the *EPG1* model calibrated with our data performed slightly worse than the *EP* models developed in present study.

There is a lack of agreement between authors regarding the exponents for metabolic weight of fish. Clarke and Johnston (1999) reported that a metabolic body weight exponent for energy of 0.75 can be used for several fish species. Ye et al. (2021) also stated that the scaling exponent for Nile tilapia is not significantly different from 0.75, as described by "Kleiber's law" (Kleiber, 1932). Some authors (Lupatsch et al., 1998, 2003b) have used measurements obtained under fasting conditions to help examine the relationship between changes in energy or protein and body weight. Ultimately, their findings indicate that protein and energy loss cannot be described by the same metabolic body weight. Others, like van Trung et al. (2011) and Glencross et al. (2011) reported for Nile tilapia and Pangasius catfish (respectively) a metabolic body weight exponent of 0.80 for both energy and protein. Both argue that it may have to do with the fact that tilapia is a low trophic level species, which results in different protein utilisation parameters compared to carnivorous species. In this study, we developed models that assumed either fixed/standard exponents for energy and protein (0.80 and 0.70, respectively) according to Lupatsch et al. (1998, 2003b, 2010) and Glencross (2008) reports, and models in which these exponents were estimated. Although the estimates of body weight exponents (0.71–0.77 for energy and 0.76–0.78 for protein) were close to standard, models in which the standard metabolic body weight exponents (0.80 and 0.70, for energy and protein, respectively) were used showed a better predictive capability than methods that considered the estimated exponents. This shows that, apparently, it is possible to use the standard values of metabolic body weight exponents (0.80 and 0.70, for energy and protein, respectively), even for omnivorous fish species.

5. Conclusion

The results of this study contributed to a better modelling of the

processes involved in Nile tilapia growth and body composition regulation and underline the importance of having a dataset covering a wide range of rearing and feeding conditions for model development. Moreover, the evaluation of models with different calibration assumptions and cross-validation methods, allowed an exploration of different possibilities and contributed to the clarification of important scientific (e.g., best model to predict growth) and technical questions (e.g., best calibration method). This study also highlighted that using information on protein intake is important to improve the estimations of Nile tilapia's growth and body composition models. Additionally, the standard body weight exponents for maintenance costs for energy and protein (0.80 and 0.70, respectively) seem to be applicable also to Nile tilapia. It can, therefore, be speculated that omnivorous fish species use energy and protein in a similar way to carnivores. Ultimately, in some cases, the use of mixed-effects modelling for model calibration can be a good approach to overcome problems related to data heterogeneity. Thus, the developed work provides a meaningful contribution to a more widespread adoption of highly predictive nutrient-based fish growth models for Nile tilapia, along with best practices in their development and evaluation, which can also be applied to other fish species.

Nevertheless, despite the good prediction quality demonstrated by the EP models developed in this work, they have some limitations. For example, the inability to predict the impact of diets with imbalanced amino acid profiles on fish growth. Therefore, in the future it is important to incorporate other mechanistic features in such growth models, besides the addition of energy and protein intake. Additionally, modelling of amino acid metabolism is likely to increase the predictive accuracy of growth models for Nile tilapia and other fish species, providing a useful tool for the aquafeed sector.

CRedit authorship contribution statement

A.I.G. Raposo: Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **F. Soares:** Conceptualization, Data curation, Investigation, Methodology, Validation, Visualization, Writing – review & editing. **A. Nobre:** Conceptualization, Writing – review & editing. **L.E.C. Conceição:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing. **L.M.P. Valente:** Funding acquisition, Project administration, Resources, Supervision, Writing – review & editing. **T.S. Silva:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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